Dual-System Architecture for Lifelong Robot Learning

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1 Architecture

We implemented a dual-system architecture using the LeRobot framework [1]. Our choices were guided by simplicity and our compute budget, prioritizing a functional prototype. For System 1, we adopted ACT [5] with a ResNet-18 [2] image backbone. This provides fast, reactive action chunking suited for high-frequency motor control. For System 2, we used a frozen CLIP [4] encoder (ViT-B/16) to extract semantic scene and language embeddings. These embeddings are injected into System 1 as a context token. Refer to Figure 1 for a visual representation of the architecture used.

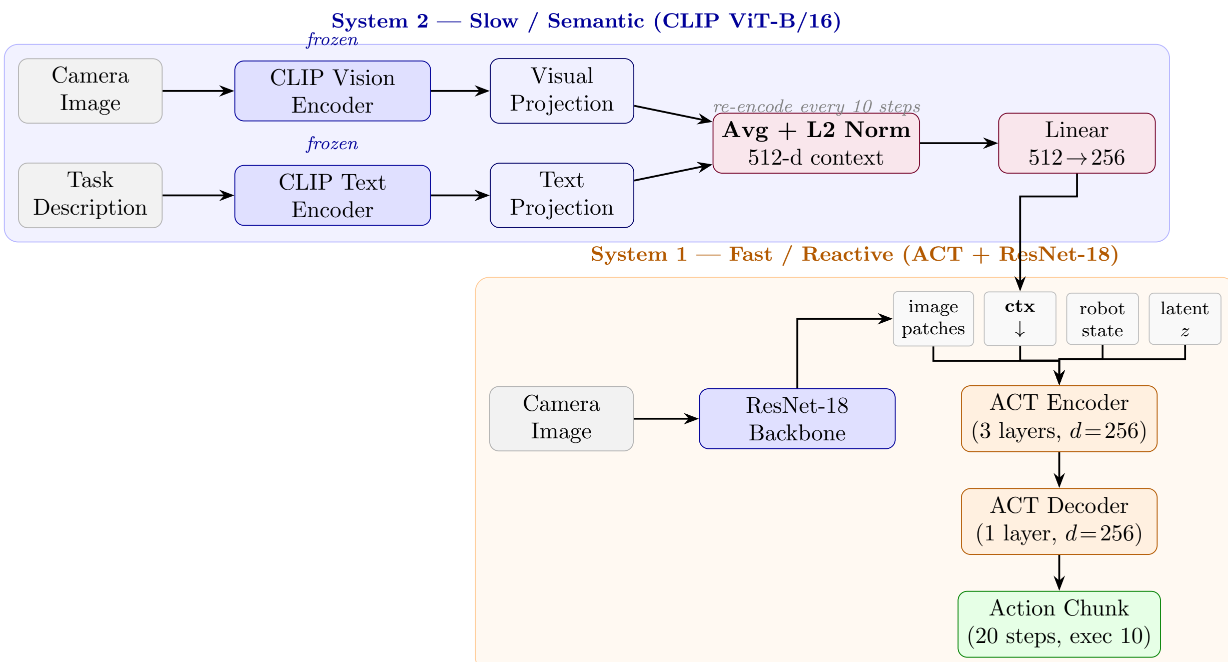

Figure 1: Dual-system architecture.

System 2 encodes the current camera image and task description with a frozen CLIP ViT-B/16, producing L2-normalised embeddings $\vec{v}, \vec{t} \in \mathbb{R}^{512}$ which are averaged into a 512-d context vector:

$$\vec{c} = \tfrac{1}{2}\left(\frac{\vec{v}}{\|\vec{v}\|} + \frac{\vec{t}}{\|\vec{t}\|}\right), \qquad \vec{c} \in \mathbb{R}^{512}. \tag{1}$$

A learned linear layer $W \in \mathbb{R}^{256\times512}$ projects this into System 1's 256-d token space, $\vec{c}' = W\vec{c} \in \mathbb{R}^{256}$, where it is prepended to the ACT encoder's input sequence alongside the VAE latent, robot state, and ResNet-18 image patch tokens. The encoder (3 layers, d=256, 4 attention heads) attends jointly over all tokens, allowing full cross-attention to the semantic context. The decoder produces a 20-step action chunk, of which 10 steps are executed before re-querying. Only W and System 1's parameters are trained; CLIP remains frozen throughout.

2 Ablation

We evaluate on the LIBERO-Spatial benchmark [3], which consists of 10 tabletop manipulation tasks requiring precise spatial reasoning (e.g., picking objects from specific locations relative to landmarks). Each task is evaluated over 20 episodes with randomised initial conditions, for a total of 200 episodes per variant.

Ablation variants. To isolate the contribution of System 2, we train two variants under identical conditions:

- **Disabled**: the System 2 context vector is replaced with a zero vector, reducing the model to a plain ACT baseline conditioned only on the current image and robot state.
- **Dynamic**: CLIP re-encodes the current scene image and task description every 10 environment steps, providing an updated semantic context throughout the episode.

All variants share the same System 1 architecture (ResNet-18 backbone, ACT transformer with d=256, 3 encoder layers, chunk size 20) and are trained for 100k steps on the `lerobot/libero_spatial_image` dataset with batch size 32. A success is recorded when the environment reward reaches 1.0 within the episode time limit. Results can be seen in Table 1.

Task	**Disabled**	**Dynamic**
Task 0	0	0
Task 1	0	0
Task 2	0	**60**
Task 3	0	10
Task 4	0	0
Task 5	0	0
Task 6	0	20
Task 7	0	0
Task 8	0	0
Task 9	0	0
Overall	**0**	**9**

Table 1: Per-task success rate (%) on LIBERO-Spatial (20 episodes per task). *Disabled*: System 2 replaced with a zero vector. *Dynamic*: System 2 re-encodes scene + language every 10 steps.

Counterfactual. Enabling System 2 raises overall success from 0% to 9%, with the largest gain on Task 2 (0% to 60%). Though absolute performance is low, the consistent edge of *Dynamic* over the zero-vector *Disabled* baseline indicates the semantic context contributes signal rather than noise.

Error recovery. A continuously re-encoding System 2 should help System 1 recover from compounding errors by re-anchoring the policy as it drifts into unseen states, whereas a static-embedding baseline cannot. Verifying this directly (ex. injecting mid-episode perturbations and comparing recovery across variants) is left to future work (Section 3).

Bottleneck analysis. Our evidence points to an *execution* rather than *reasoning* bottleneck. System 2 is a frozen, pretrained CLIP encoder on in-distribution scenes and templated language, so it is unlikely to be limiting. Meanwhile, success is near zero on most tasks even with *Dynamic* enabled, yet Task 2 reaches 60%—showing System 1 *can* execute when training suffices. Broad failure is hypothesized to be with undertrained control than semantic misguidance.

3 Limitations

Compute was the primary bottleneck and shaped most of our design decisions. It limited both the number of ablation variants and total training duration. With more resources we would evaluate a *Frozen* variant (System 2 encoding once at episode reset and holding the context fixed) to isolate the value of continuous re-encoding against a one-shot semantic prior.

The compute budget capped training at 100k steps, likely insufficient for ACT to converge given the broadly low success rates (Section 2). We would extend training to roughly 300k steps and monitor validation success before drawing firm conclusions about System 2's contribution.

The same budget precluded the perturbation-based error-recovery and oracle-context reasoning experiments. These are most informative once baseline execution improves, since at current success rates they would largely measure System 1 failures. Architectural choices (ACT + ResNet-18 for System 1, frozen CLIP for System 2) were driven by availability of pretrained weights and compatibility with the LeRobot pipeline.

References

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